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PNT Lab Internship – Final Report

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Road Surface Condition Mapping Using Computer Vision and GNSS

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1. Introduction

This internship under the **PNT Lab** focused on applying positioning and geospatial technologies to a real-world problem. The purpose of the project was to develop a **road surface condition mapping system** that detects potholes and cracks using computer vision and geo-tags their locations using GNSS. The project scope includes defect detection, location logging, and visualization through GIS heatmaps to support road maintenance planning.

The work is relevant to **geospatial and positioning technologies** as it integrates GNSS for accurate location identification and GIS for spatial data analysis. The project demonstrates a practical application of PNT technologies in infrastructure monitoring and smart transportation systems.

2. Aim and Objectives

Aim

The aim of this project is to develop an automated system for **road surface condition mapping** using computer vision and GNSS to detect road defects and visualize their locations for effective road maintenance planning.

Objectives

- To capture road surface images and videos using a vehicle-mounted camera.
- To detect potholes and cracks using computer vision techniques.
- To geo-tag detected road defects using GNSS coordinates.
- To store defect data with time and location information in CSV format.
- To generate GIS heatmaps for visualizing road condition severity.
- To support proactive decision-making for municipal road maintenance.

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3. Methodology

The project was carried out through a structured workflow involving data collection, processing, analysis, and visualization of road surface conditions.

Data Collection

- Mounted a camera on a moving vehicle using a Raspberry Pi or Android smartphone.
- Captured continuous road surface images and videos during vehicle movement.
- Used a GNSS receiver module (NEO-M8T) to record latitude, longitude, and timestamp data.
- Collected data on selected local roads under real driving conditions.

Software and Technologies Used

- Python for image processing and data handling.
- OpenCV for frame extraction and road defect detection.
- Pre-trained deep learning model (YOLO/CNN) for pothole and crack detection.
- Pandas for organizing and storing detection results in CSV format.
- QGIS for importing data and generating road condition heatmaps.

Analytical and Modeling Methods

- Processed captured video frames using computer vision techniques.
- Trained and tested the detection model using labeled road images.
- Matched detection timestamps with GNSS data to geo-tag each defect.
- Tuned model parameters to reduce false detections and improve accuracy.

Experimental Setup / Workflow

- Vehicle-mounted camera captures road footage.
- Computer vision model detects potholes and cracks from video frames.

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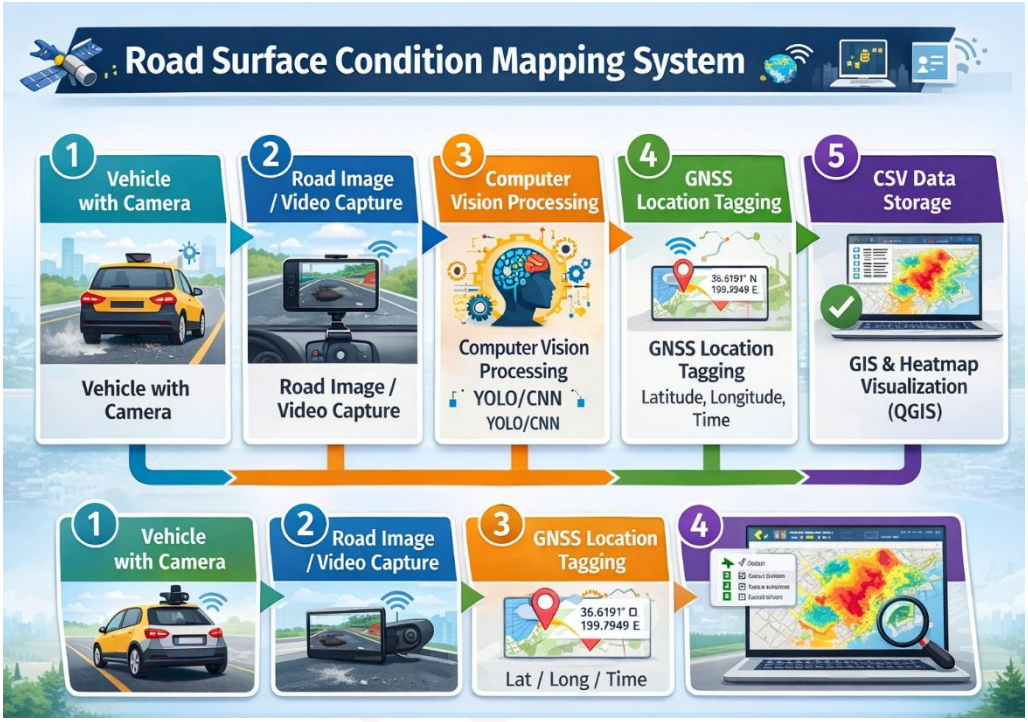
- GNSS module records location data for each detected defect.
- Detection and location data stored in CSV format.
- GIS heatmaps generated to visualize road condition severity.



4. Results and Discussion

Results

- The developed system successfully detected **road surface defects such as potholes and cracks** from captured road images and videos.
- Detected defects were accurately **geo-tagged using GNSS coordinates** (latitude, longitude, and timestamp).
- The detection results were stored in **CSV format**, containing defect type, location, and time of detection.
- The CSV data was imported into **QGIS**, and **heatmaps were generated** to visualize road condition severity.
- Heatmaps clearly highlighted **high-risk road segments**, enabling easy identification of damaged road areas.
- Screenshots of detected potholes, CSV logs, and GIS heatmaps were generated as visual outputs.

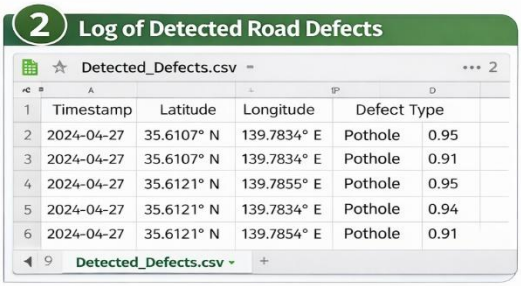
Discussion

- The computer vision model demonstrated effective performance in identifying visible potholes and cracks under normal lighting conditions.
- GNSS-based geo-tagging enabled accurate spatial mapping of defects, making the system suitable for geospatial analysis.
- GIS heatmaps provided an intuitive visualization of road conditions, which can assist municipal authorities in prioritizing road maintenance.
- Some false detections were observed due to shadows, road markings, or lighting variations; these were reduced through parameter tuning and preprocessing.
- The system proved to be **cost-effective, scalable, and practical** for real-world road monitoring applications.
- Overall, the project validates the usefulness of integrating **computer vision, GNSS, and GIS** for smart infrastructure monitoring.

Key Insights

- Automated road inspection reduces manual effort and inspection time.
- Spatial visualization improves decision-making for maintenance planning.
- Integration of PNT technologies enhances accuracy and reliability.
- The system can be extended for real-time monitoring and larger road networks.

Fig. 1. Pothole Detection Results



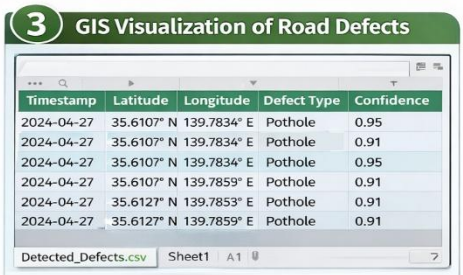


Fig. 1. Log of Detected Road Defects

Fig. 3. GIS Visualization of Road Defects

5. Conclusion

This project successfully demonstrated an automated approach for **road surface condition mapping** using computer vision and GNSS technologies. The developed system was able to detect potholes and cracks from road images, accurately geo-tag their locations, and visualize the results using GIS heatmaps. The outcomes show that integrating computer vision with positioning and geospatial technologies can significantly improve road condition monitoring and support proactive maintenance planning. The project highlights a practical and cost-effective solution that can enhance road safety and infrastructure management.

The internship under the **PNT Lab** greatly contributed to the development of technical and research skills. It provided hands-on experience in computer vision, GNSS data handling, GIS visualization, and real-world data analysis. Additionally, the internship improved problem-solving abilities, system integration skills, and understanding of geospatial applications, preparing a strong foundation for future work in positioning, navigation, and smart infrastructure technologies.

6. Learning Experience

Technical Skills

During the internship, I gained hands-on experience in computer vision and geospatial technologies. I learned to use **Python and OpenCV** for image and video processing, implement **deep learning models (YOLO/CNN)** for defect detection, and handle **GNSS data** for accurate location tagging. I also developed skills in **data handling using CSV and Pandas**, and **GIS visualization using QGIS** for generating heatmaps and spatial analysis.

Soft Skills

The internship helped improve essential soft skills such as **communication and teamwork** through regular discussions with mentors and peers. I learned to manage project timelines, plan tasks effectively, and document progress systematically. Problem-solving and critical thinking skills were enhanced while debugging models and refining detection accuracy.

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Exposure to PNT Lab Ecosystem

The internship provided valuable exposure to the **PNT Lab ecosystem at Geo-Intel Lab, IITTNiF**. Interactions with experienced mentors helped in understanding real-world applications of positioning, navigation, and timing technologies. The experience offered insight into research-oriented project development, interdisciplinary collaboration, and practical implementation of geospatial solutions.

7. Gratitude and Acknowledgement

I would like to express my heartfelt gratitude to Geo-Intel Lab, IIT Tirupati Navavishkar I-Hub Foundation (IITTNiF) for providing this opportunity through the PNT Lab hosted at our institution.
The mentorship, resources, and collaborative ecosystem created by the Geo-Intel Lab, IITTNiF, have been instrumental in enhancing my understanding of geospatial and positioning technologies.
I am thankful to my faculty mentor, peers, and IITTNiF team for their continuous guidance and encouragement throughout this internship.

8. References

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